DIGITAL FORENSIC AND SOC AUDITING (MODULE 1)

Unit 1: Fundamentals Of Digital Forensics

Priyanka Sharma

DIGITAL FORENSICS

Digital Forensics is the process of uncovering, preserving, acquiring, analysing, and presenting digital evidence in a legally admissible manner. It applies scientific techniques to digital devices to reconstruct past events during an investigation.

DIGITAL FORENSICS Science / Technology + Legal Standards = Admissible Evidence

Core Objectives

Preserve Evidence Integrity: Ensure evidence remains unaltered from seizure to court.

Identify Cause and Intent: Determine who, what, when, where, and how an incident occurred.

Reconstruct Timelines: Rebuild the chronological sequence of malicious activities or system events.

Maintain Chain of Custody: Document every individual who handles the evidence.

Ensure Legal Admissibility: Produce evidence that withstands courtroom scrutiny.

DIGITAL EVIDENCE

Digital evidence (also known as electronic evidence) is any binary data or information stored, transmitted, or processed by an electronic device that can be relied upon in court or an internal investigation to prove or disprove a fact.

According to **ISO/IEC 27037**, digital evidence must maintain its probative value (ability to prove a point) through rigorous acquisition, handling, and analysis practices.

Key Characteristics of Digital Evidence

Unlike physical evidence (like fingerprints or blood samples), digital evidence has unique properties that require specialized handling:

Fragile and Volatile: Can be easily altered, overwritten, or destroyed by simple actions (e.g., booting up a computer changes system timestamps and overwrites RAM).

Latent (Hidden): Data is not directly visible to the naked eye. It requires specialized software to extract metadata, deleted files, or raw unallocated space.

Easily Duplicable: Exact 1:1copies (bit-stream images) can be created without degrading the original source material.

Time-Sensitive: Transient network connections and volatile memory (RAM) vanish as soon as power is cut or overwritten by system processes.

DIGITAL EVIDENCE

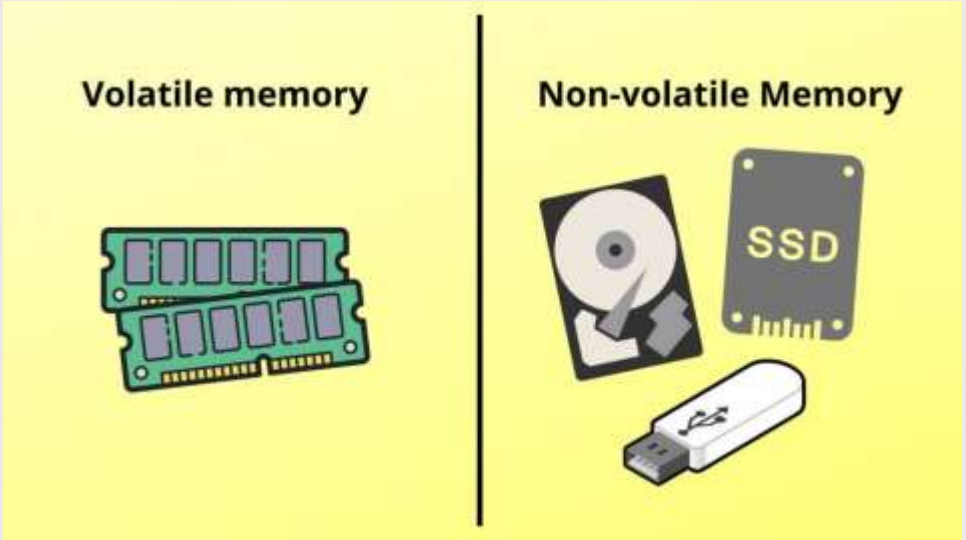
According to Locard's Exchange Principle, "**anyone or anything, entering a crime scene takes something of the scene, and leaves something of themselves behind**."



TYPES OF DIGITAL EVIDENCE

- **Volatile Data:** Volatile data refers to the temporary information on a digital device that requires a constant power supply and is deleted if the power supply is interrupted.
- E.g., system time, logged-on user(s), open files, network information, process information, process-to-port mapping, process memory, clipboard contents, service/driver information, command history, etc.

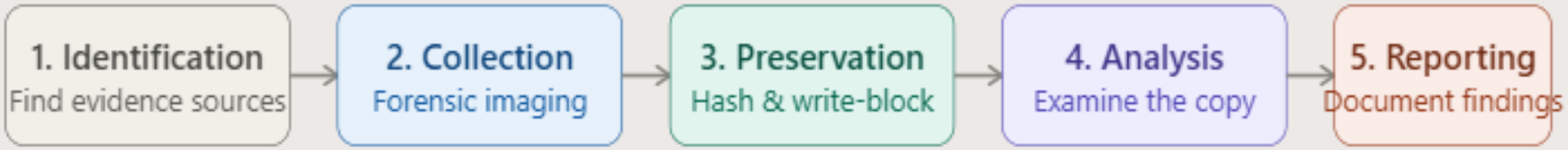
- **Non-volatile Data:** Non-volatile data refers to the permanent data stored on secondary storage devices, such as hard disks and memory cards.
- E.g., hidden files, slack space, swap file, index.dat files, unallocated clusters, unused partitions, registry settings, and event logs.



MAJOR TYPES OF DIGITAL FORENSICS



THE 5-PHASE FORENSIC PROCESS



Golden rule: always work on a forensic copy — the original evidence is never modified

DIGITAL FORENSIC PROCESS

1. Identification

Recognize potential sources of evidence: computers, servers, mobile devices, cloud accounts, IoT devices, removable media.

Determine the scope of the investigation — what data is relevant to the incident or crime.

Assess volatility: some data (RAM, network sessions) disappears if the system is powered off, so it must be prioritized.

Secure the scene to prevent evidence from being altered before collection begins.

2. Collection

Acquire data using forensically sound methods — typically a **bit-by-bit (bitstream) image** of the storage media, not a simple file copy.

Obtain proper legal authorization first: a warrant, documented consent, or an exigent circumstance.

Use write-blockers to physically or logically prevent any write operation to the original media during imaging.

Begin the chain-of-custody log the moment evidence is touched — who collected it, from where, and when.

3. Preservation

Protect the collected evidence from alteration, damage, or deletion for the duration of the case.

Generate cryptographic hash values (MD5, SHA-1, SHA-256) of the image immediately after acquisition — this hash becomes the fingerprint that proves the copy hasn't changed later.

Store the original media and its verified image in secure, access-controlled, logged storage.

Re-verify hashes at each stage the evidence is accessed, to demonstrate continued integrity.

4. Analysis

- Work only on the verified copy/image — the original is never touched again after acquisition.
- Recover deleted files, examine file system metadata, registry entries, logs, and browser/app artifacts.
- Reconstruct timelines by correlating timestamps across multiple sources.
- Cross-reference findings — e.g., linking a network log entry to a specific file accessed on a device.
- Use validated, court-accepted tools (EnCase, FTK, Autopsy, etc.) so the methodology itself can withstand scrutiny.

5. Reporting

- Document findings in a clear, structured, and objective report — methodology, tools used, evidence handled, and conclusions.
- Write for two audiences at once: technically precise enough for peer review, but understandable to judges, juries, or executives.
- Include the chain-of-custody trail and hash verifications as part of the evidentiary record.
- Be prepared to defend the findings under cross-examination if the case goes to trial.

Golden rule across all five stages: never work on the original evidence — every action is performed on a verified forensic copy, and every action taken is logged.

CHECKLIST TO PREPARE FOR A COMPUTER FORENSICS INVESTIGATION

- Do not turn the computer off or on, run any programs, or attempt to access data on the computer.
- Secure any relevant media including hard drives, cell phones, DVDs, USB drives, etc subject may have used
- Suspend document destruction and recycling that may pertain to relevant media or users at the time of issue
- Perform a preliminary assessment of the crime scene and identify the type of data you are seeking, the information you are looking for, and the urgency level of the examination
- Once the machine is secured, obtain info about the machine, the peripherals, and network where connected
- If possible, obtain passwords to access encrypted or password-protected files
- Compile a list of names, e-mails, and other info of those with whom the subject might have communicated
- If the computer is accessed before the forensic expert is able to secure a mirror image, note the user(s) who accessed it, what files accessed, and when access occurred. If possible, find out why the pc was accessed
- Maintain a chain of custody for each piece of original media, indicating where the media has been, whose possession it has been in, and the reason for that possession. 10 Create a list of key words or phrases to use when searching for relevant data

CHAIN OF CUSTODY

- Legal document that demonstrates the progression of evidence as it travels from the original evidence location to the forensic laboratory.
- It is a roadmap that shows how investigators collected, analyzed, and preserved the evidence.
- It ensures accurate auditing of the original data evidence, imaging of the source media, tracking of the logs, and so on.
- Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA) governs the admissibility of electronic records in Indian courts, replacing the old Section 65B of the Indian Evidence Act. It permits computer and digital output—such as emails, WhatsApp chats, and CCTV footage—to be admitted as evidence without producing the original device, provided it is accompanied by a mandatory certificate.

Single Evidence Form

Case No. | Evidence No.

Digital Forensics Lab

PLEASE COMPLETE FORM IN UPPERCASE

Section B: Evidence Collection

Date/Time Collected | Collected by

Site Address

Section C: Evidence Details

Date/Time Stored

Storage Location

Device Type | Capacity

Manufacturer | Model

Serial No.

MD5 Sum

SHA-1 Sum

Additional Information...

Note any damage, marks and scratches | Digital Image Taken ☐ Yes ☐ No

Section D: Image Details

Date/Time Imaged | Imaged by

Storage Location

Image Filename | Image Size (inc. unit)

Additional Information...

This form is to be used when collecting a hardware device containing data that may be of interest in a case. Guidelines:

- **Ensure that this form only refers to one item of evidence and that one is completed for each item of evidence**
- **This form must be accompanied by Chain of Custody forms which detail the individuals that have handled the evidence**
- **Further remarks can be noted overleaf in Section E: Remarks**
- **It is important that these forms are kept with the evidence at all times**
- **Upon handover or disposal please complete Section F: Evidence Handover**

Chain of Custody Form

for use with a Single Evidence form

Case No. | Evidence No. | Page No.

Digital Forensics Lab

This form must accompany a Single Evidence form and it's respective evidence

Chain of Custody

SUBMITTER		RECEIVER
Name:		Name:
Signature:		Signature:
	Evidence Modified:	
Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
Signature:		Signature:
	Evidence Modified:	
Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
Signature:		Signature:
	Evidence Modified:	
Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
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Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
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Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
Signature:		Signature:
	Evidence Modified:	
Date & Time:	Yes / No	Date & Time:
SUBMITTER		**RECEIVER**
Name:		Name:
Signature:		Signature:
	Evidence Modified:	
Date & Time:	Yes / No	Date & Time:

If this form is full please continue on another page

ROLE OF INVESTIGATORS

A forensic investigator performs the following tasks:

- Evaluates the damages of a security breach
- Identifies and recovers data required for investigation
- Extracts the evidence in a forensically sound manner
- Ensures proper handling of the evidence
- Acts as a guide to the investigation team
- Creates reports and documents about the investigation required to present in the court of law.
- Reconstructs the damaged storage devices and uncovers the information hidden on the computer.
- Updates the organization about various methods of attack and data recovery techniques and maintains a record of them (following a variant of methods to document) regularly.
- Addresses the issue in a court of law and attempts to win the case by testifying in court.

CYBER LAWS

Cyber laws are the legal rules and regulations that govern the use of computers, digital devices, networks, the Internet, and electronic communications. These laws are designed to prevent cybercrimes, protect digital information, safeguard individual privacy, regulate electronic transactions, and ensure the secure use of information technology.

Objectives of Cyber Laws

The primary objectives of cyber laws are:

To protect computer systems, networks, and digital information from unauthorized access and misuse.

To provide legal recognition to electronic records and digital signatures.

To prevent and punish cybercrimes.

To ensure the privacy and security of individuals and organizations.

To facilitate secure electronic commerce and online banking.

To establish procedures for the investigation and prosecution of cyber offenses.

To promote responsible and ethical use of information technology.

INFORMATION TECHNOLOGY (IT) ACT, 2000

The **Information Technology Act, 2000** is the primary cyber law in India. It came into force on **17 October 2000** and was enacted to provide a legal framework for electronic governance, digital transactions, and cybercrime prevention. The Act was significantly amended in **2008** to address emerging cyber threats and technological advancements.

Important Provisions of the IT Act, 2000

Some of the key sections of the Information Technology Act include:

Section 43: Provides compensation for unauthorized access, downloading, introducing viruses, damaging computer systems, or disrupting services.

Section 65: Punishes tampering with computer source code.

Section 66: Deals with computer-related offenses such as hacking and unauthorized access performed dishonestly or fraudulently.

Section 66C: Addresses identity theft involving passwords, digital signatures, or other unique identification features.

Section 66D: Covers cheating by personation using computer resources, including online fraud and phishing.

Section 66E: Punishes violations of privacy through the capture or publication of private images without consent.

Section 67: Deals with publishing or transmitting obscene material in electronic form.

Section 69: Grants the Government the authority to intercept, monitor, or decrypt electronic information under specified legal conditions.

Section 72: Provides punishment for breach of confidentiality and privacy by individuals who have gained access to electronic information through legal authority.

INFORMATION TECHNOLOGY (AMENDMENT) ACT, 2008 (ITAA 2008)

The **Information Technology (Amendment) Act, 2008 (ITAA 2008)** was enacted to strengthen India's cyber laws by addressing emerging cybercrimes, promoting electronic governance, enhancing data protection, and improving cybersecurity.

◆ Structural Change	📜 Section(s)	📝 Key Impact
Electronic Signatures	Section 3A	Introduced Electronic Signatures in addition to Digital Signatures, enabling secure electronic authentication using multiple technologies.
Electronic Contracts	Section 10A	Granted legal recognition to electronic contracts executed through emails, websites, and online platforms, promoting e-commerce and e-governance.
Data Protection	Section 43A	Organizations handling sensitive personal data are liable to pay compensation if negligence leads to a data breach or security failure.
Privacy Protection	Section 72A	Punishes unauthorized disclosure of personal information obtained through lawful contracts, protecting individuals' privacy.
New Cyber Crimes	Sections 66B–66F, 67A, 67B	Added offences including identity theft, online impersonation, cyber terrorism, violation of privacy, publishing sexually explicit material, and child sexual abuse material.

National Cyber Incident Response (CERT-In)	Section 70B	Designated CERT-In as India's national agency for cybersecurity incident response, threat alerts, vulnerability reporting, and coordination.
Safe Harbour for Intermediaries	Section 79	Provides legal immunity to intermediaries (ISPs, social media platforms, web-hosting providers) if they exercise due diligence and comply with legal requirements.
Government Surveillance & Website Blocking	Sections 69A & 69B	Empowered the Government to intercept, monitor, decrypt communications, and block online content for national security and public order.
Critical Information Infrastructure Protection	Section 70A	Established a framework to identify and secure critical sectors such as banking, power, telecom, defence, and transportation from cyber attacks.

GDPR (GENERAL DATA PROTECTION REGULATION) – EUROPEAN UNION (EU), 2018

The **General Data Protection Regulation (GDPR)** is one of the world's most comprehensive data protection laws. It regulates how organizations collect, store, process, and share the personal data of individuals in the European Union. GDPR applies to any organization that processes the personal data of EU residents, regardless of where the organization is located.

Key Points:

Protects the personal data and privacy of EU citizens.

Requires organizations to obtain clear user consent before collecting data.

Grants individuals rights such as access, correction, deletion ("Right to be Forgotten"), and data portability.

Requires reporting of certain data breaches within **72 hours**.

Imposes significant financial penalties for non-compliance.

HIPAA (HEALTH INSURANCE PORTABILITY AND ACCOUNTABILITY ACT) – UNITED STATES, 1996

The **Health Insurance Portability and Accountability Act (HIPAA)** is a U.S. law designed to protect sensitive healthcare information. It establishes national standards for the privacy and security of **Protected Health Information (PHI)** and requires healthcare organizations to implement safeguards against unauthorized access and disclosure.

Key Points:

Protects patients' medical records and health information.

Applies to healthcare providers, health plans, and business associates.

Includes the **Privacy Rule**, **Security Rule**, and **Breach Notification Rule**.

Requires administrative, physical, and technical safeguards for electronic health records.

Violations can result in substantial civil and criminal penalties.

CCPA (CALIFORNIA CONSUMER PRIVACY ACT) – CALIFORNIA, USA, 2020

The **California Consumer Privacy Act (CCPA)** gives California residents greater control over how businesses collect and use their personal information. It increases transparency and requires businesses to disclose the types of data they collect and how the data is used or shared.

Key Points:

Gives consumers the right to know what personal information is collected.

Allows consumers to request deletion of their personal data.

Provides the right to opt out of the sale or sharing of personal information.

Requires businesses to provide clear privacy notices.

Protects consumers from discrimination for exercising their privacy rights.

BUDAPEST CONVENTION ON CYBERCRIME – COUNCIL OF EUROPE, 2001

The **Budapest Convention on Cybercrime** is the first international treaty specifically aimed at combating cybercrime. It promotes international cooperation by harmonizing cybercrime laws, improving investigative techniques, and facilitating the sharing of electronic evidence across countries.

Key Points:

First international treaty focused on cybercrime.

Standardizes laws related to computer crimes among participating countries.

Encourages international cooperation in cybercrime investigations.

Supports the collection, preservation, and exchange of digital evidence.

Covers offenses such as illegal access, data interference, system interference, computer fraud, and online child exploitation.

WHAT IS DIGITAL EVIDENCE?

Digital evidence is any information stored or transmitted in digital form that can be used to investigate or establish facts related to an incident or crime.

Example: If an employee copies confidential files to a USB drive, investigators may find:

USB device → File → Metadata → Timestamp → User activity → Evidence

Key Characteristics of Digital Evidence

Unlike physical evidence (like fingerprints or blood samples), digital evidence has unique properties that require specialized handling:

Fragile and Volatile: Can be easily altered, overwritten, or destroyed by simple actions (e.g., booting up a computer changes system timestamps and overwrites RAM).

Latent (Hidden): Data is not directly visible to the naked eye. It requires specialized software to extract metadata, deleted files, or raw unallocated space.

Easily Duplicable: Exact 1: 1copies (bit-stream images) can be created without degrading the original source material.

Time-Sensitive: Transient network connections and volatile memory (RAM) vanish as soon as power is cut or overwritten by system processes.

DIGITAL EVIDENCE LIFECYCLE

A typical forensic investigation follows:

Identification → Collection → Preservation → Examination → Analysis → Documentation → Presentation

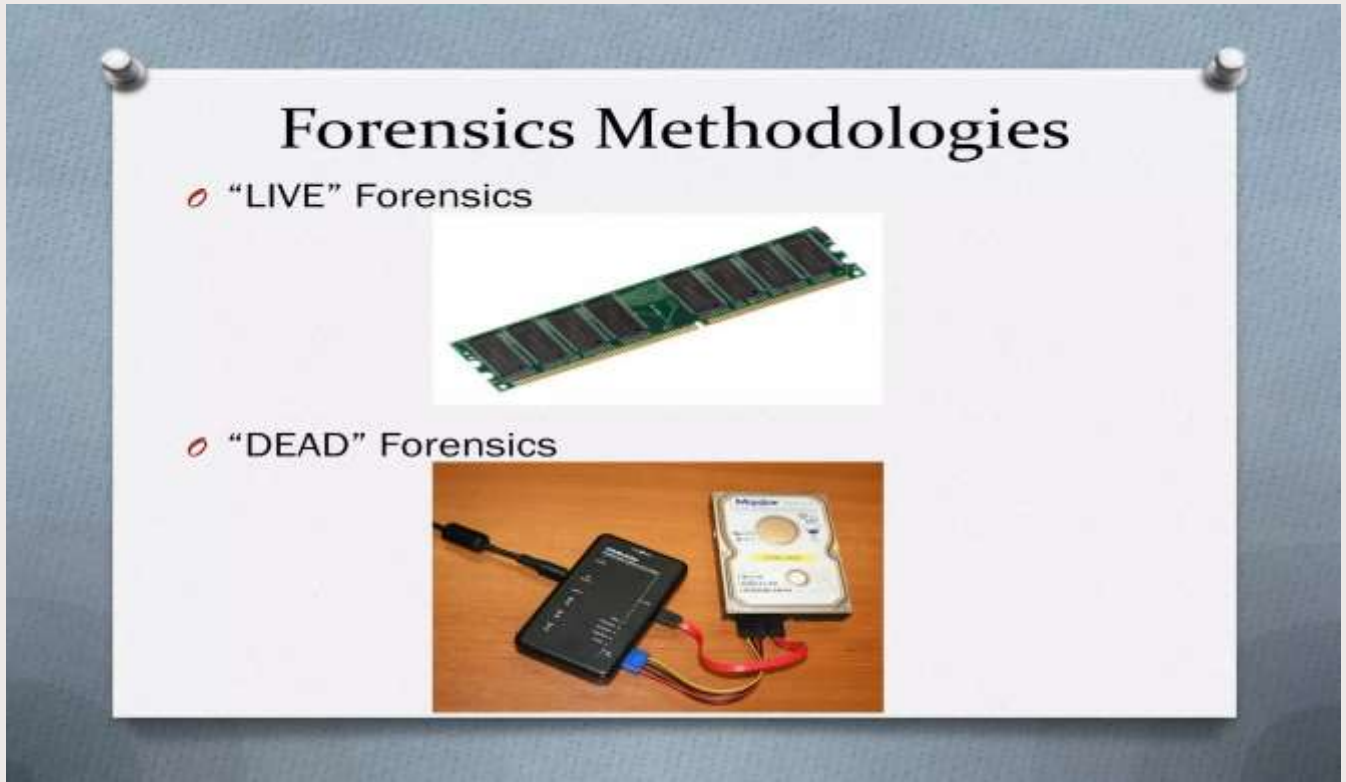


EVIDENCE COLLECTION

Evidence collection means:

Obtaining digital information from a device while maintaining its **original state and integrity**.

Potential evidence sources: Computer ,Laptop ,Mobile phone,HDD/SSD ,USB ,SD card ,Cloud account ,Network device ,Server ,IoT device.



EVIDENCE COLLECTION METHODS

- **Live Acquisition**

Evidence is collected while the system is running.

Useful for: RAM ,Running processes ,Network connections ,Encryption keys ,Active sessions

- **Dead/Offline Acquisition**

The device is powered off and storage media is acquired separately. Example:

Computer → Remove HDD → Connect using Write Blocker → Create forensic image

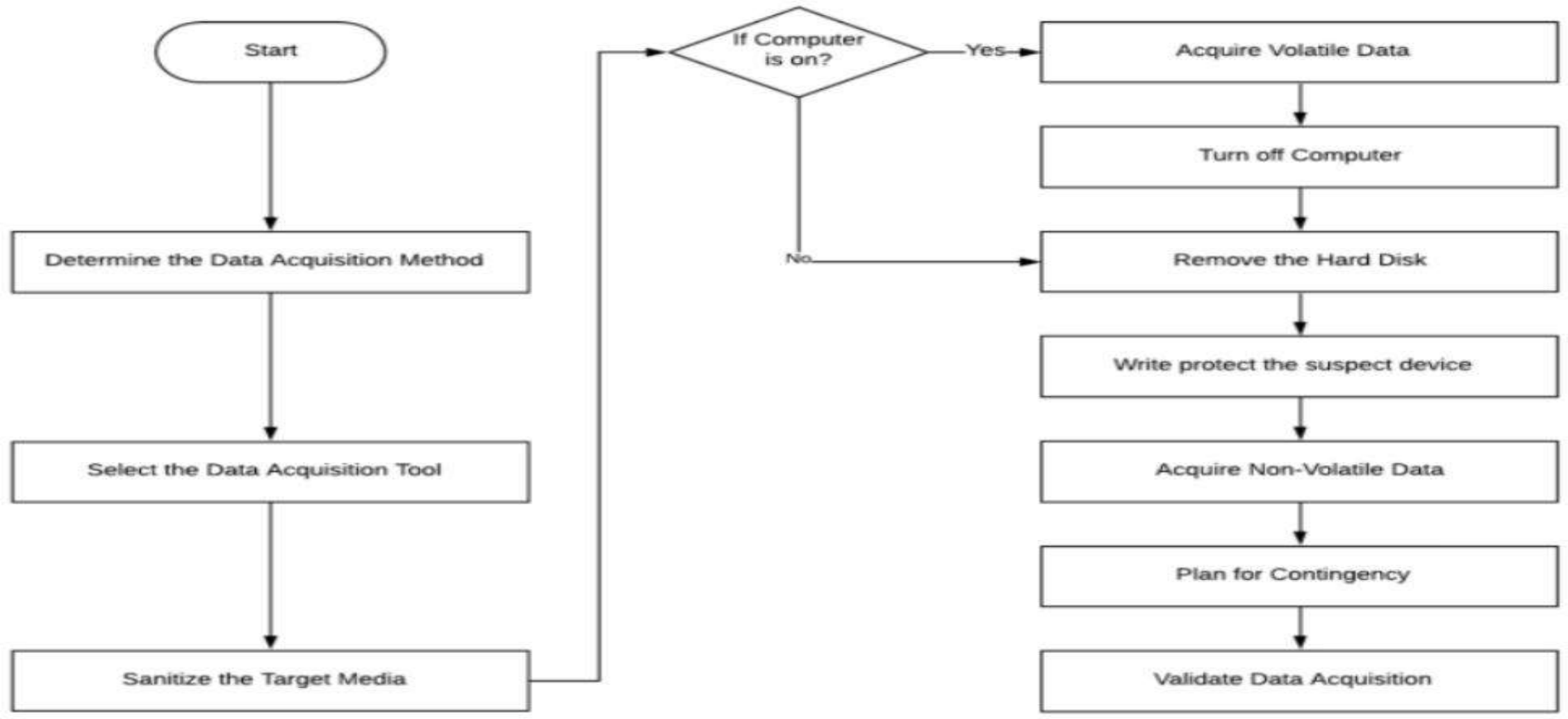
Advantages

Minimizes modification of original media.

Easier to preserve evidence integrity.

- **Physical Acquisition** – Create a bit-by-bit copy of the entire storage device.
- **Logical Acquisition** – Collect selected files, folders, or user data.
- **Network Acquisition** – Collect network packets and logs.
  Example: PCAP, firewall logs.
- **Cloud Acquisition** – Collect data from cloud services.
  Example: Emails, cloud files, login records.

EVIDENCE COLLECTION METHOD



WRITE BLOCKER

What is a Write Blocker?

A hardware or software mechanism that prevents data from being written to the original evidence storage device.

Without write protection, the forensic workstation could unintentionally modify: File timestamps,File system metadata ,System files ,Access information.



DUPLICATION OF DATA

Disk Imaging

Creates an **image file** of the disk

Example: disk.img, .E01

Can be stored on a computer/server

Commonly used in **digital forensics**

Original evidence remains separate

Easy to create multiple working copies from one image

Disk Cloning

Creates a **direct copy onto another disk**

HDD → HDD

Requires another physical storage device

Commonly used for disk duplication/deployment

Target becomes a copy of the source

Usually creates one direct disk copy

INTEGRITY VERIFICATION

Integrity verification is the process of ensuring that digital evidence has **not been altered, modified, or corrupted** during acquisition, examination, or storage.

Why is integrity important?

Maintains the **authenticity** of digital evidence

Detects accidental or intentional modifications

Ensures the evidence presented in court is the **same evidence that was originally collected**

Supports the **chain of custody**

Helps establish forensic reliability

Basic principle:

Original Evidence → Hash Calculation → Examination → Hash Recalculation → Comparison

If the hashes match → **Evidence is considered unchanged.**

If the hashes differ → **Evidence may have been modified or corrupted.**

HASHING

A **hash function** converts digital data of any size into a fixed-length value called a **hash value, hash digest, or message digest**.

Common Hashing Algorithms:

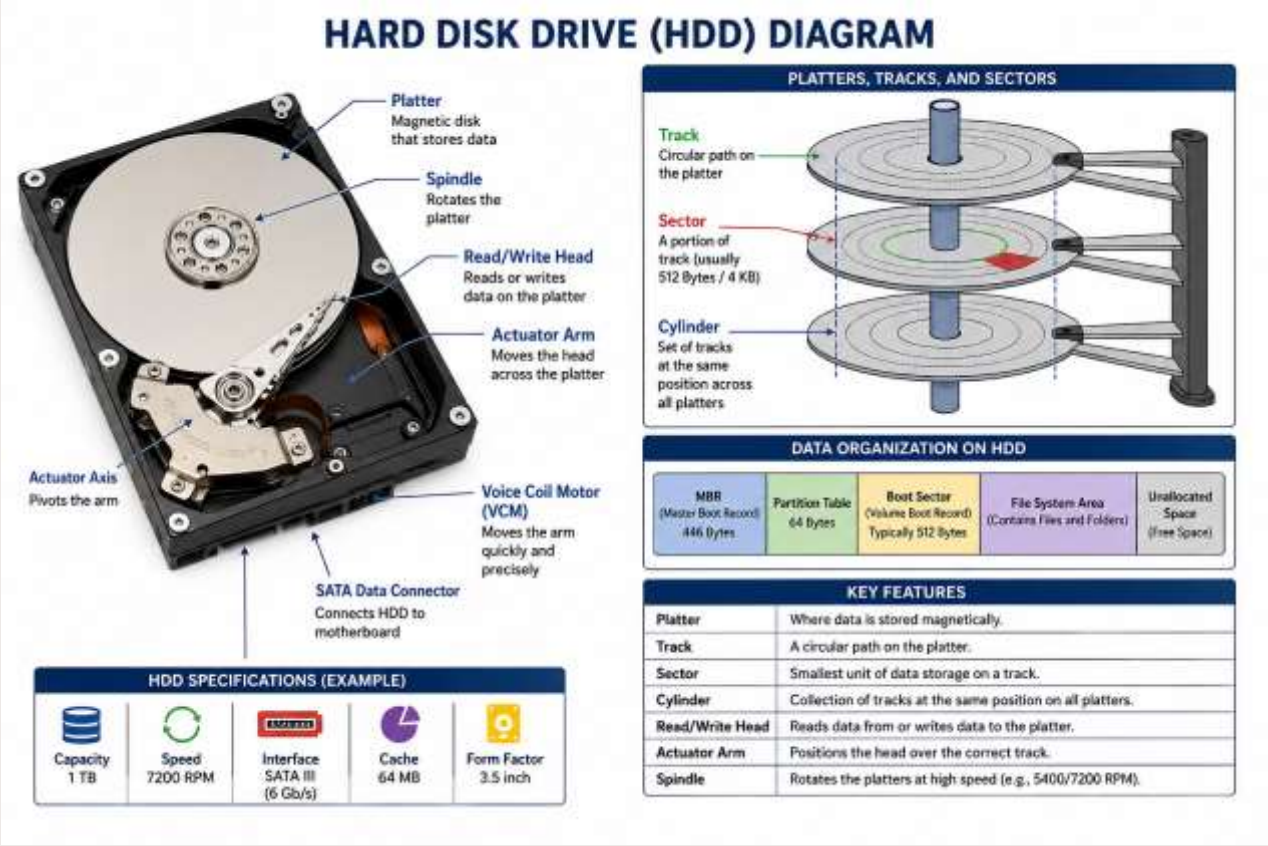
Algorithm	Hash Length	Current Forensic Use
MD5	128 bits / 32 hex characters	Legacy; useful for compatibility, not ideal as sole integrity proof
SHA-1	160 bits / 40 hex characters	Legacy; collision attacks make it unsuitable as a modern primary integrity algorithm
SHA-256	256 bits / 64 hex characters	**Preferred modern choice**

GLA University
'A+' Grade by NAAC | 12-B Status from UGC

Data Recovery & Storage Forensics

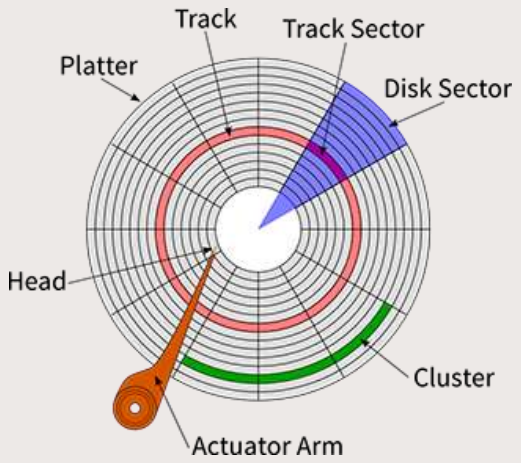
UNDERSTANDING HARD DISK DRIVE

- HDD is a non-volatile digital data storage device that records data magnetically on a metallic platter
- he read/write performance of an HDD is directly proportional to the RPM (revolutions per minute) of the drive platter
- Since an HDD contains moving parts, it is susceptible to physical damage, and the moving parts wear out over the time



HARD DISK DRIVE PARTS

- **Platters** circular metal disks mounted into a drive enclosure
- **Tracks** are concentric rings on the platters that store data; each track has smaller partitions called disk blocks or sectors. Track numbering starts at 0.
- **Sectors** are the smallest physical storage units located on a hard disk platter and are 512bytes long. New format sectors are 8, 512B sectors to make one 4096B or 4KB sector; which is more efficient
- **Clusters** are the smallest accessible/logical storage units on the hard disk. Clusters form by combining sectors in order to ease the process of handling files. Also called allocation units, the clusters are sets of tracks and sectors ranging from 2 to 32, or more, depending on the formatting scheme.
- **Cylinder** is a group of tracks that are located at the same position on each of the disk surfaces



CHS (CYLINDER-HEAD-SECTOR) DATA ADDRESSING AND DISK CAPACITY CALCULATION

The CHS addressing method addresses each physical block of data on a hard disk by specifying the cylinder (radius), head (platter side), and sector (angular position).

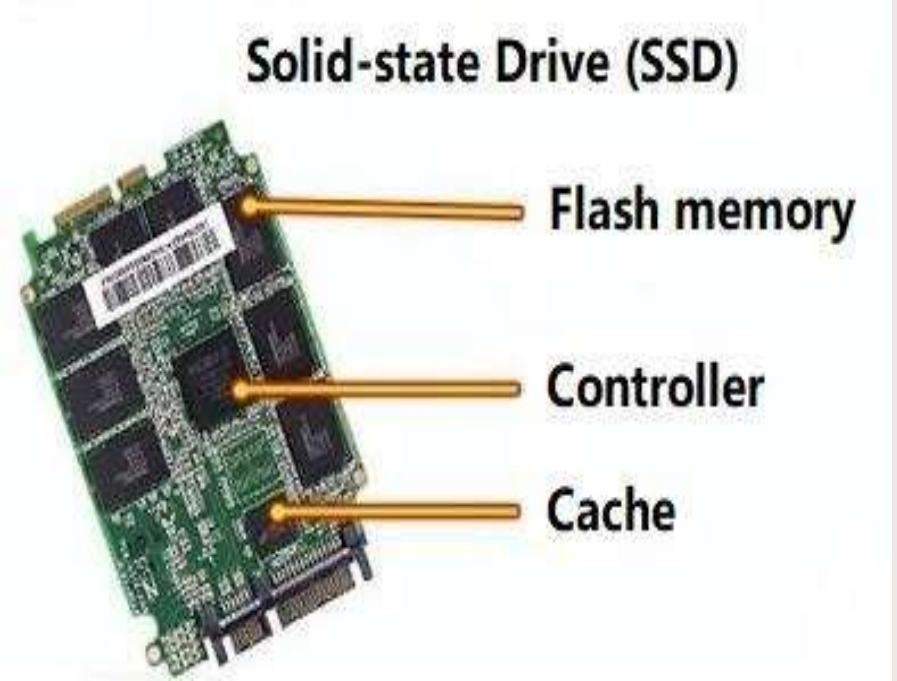
E.g., A disk drive has 16,384 cylinders, 80 heads, and 63 sectors per track. Assume - a sector has 512 bytes. What is the capacity of such a disk?

Total Size of the Disk = No. of Cylinders * No. of Heads * No. of Sectors per Track * 512 bytes per Sector Total Size of the Disk = (16,384 cylinders) * (80 heads) * (63 sectors / track) * (512 bytes / sector) = 42,278,584,320 bytes

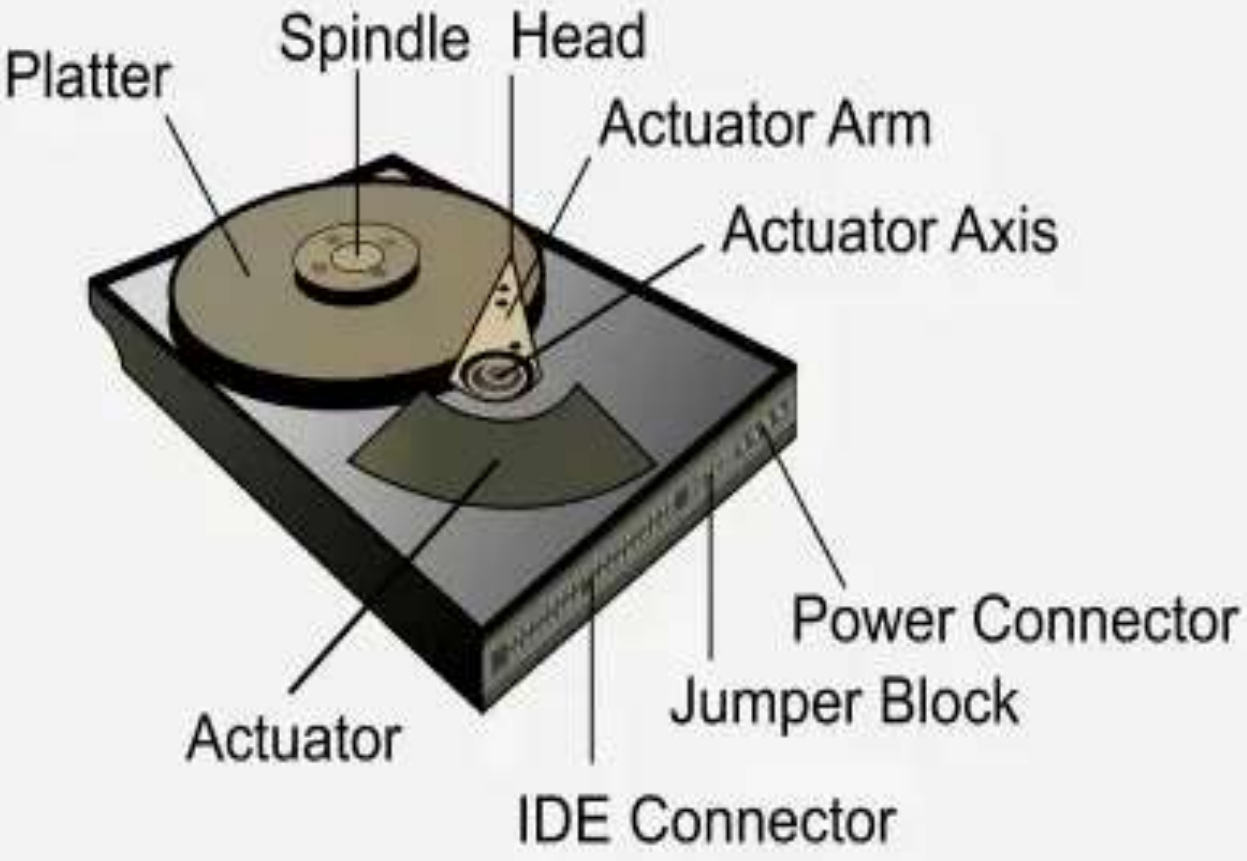
UNDERSTANDING SOLID-STATE DRIVE (SSD)

- SSD is a non-volatile storage device that uses NAND flash memory chips to store digital data
- SSDs are faster than HDDs as they have no moving parts, and the read/write performance depends on data connection of the drive

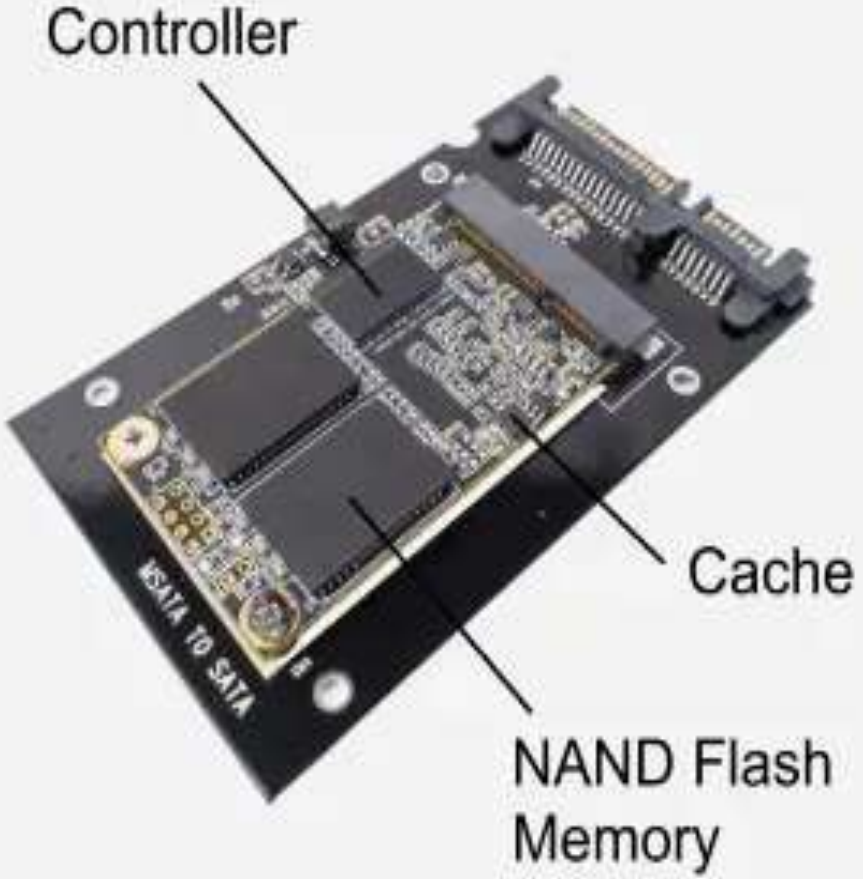
- NAND Flash Memory: It is the main data storage unit made up of floating gate transistors which retain the charge state even without power
- Controller: It is a processor that acts as a bridge between the flash memory components and the computer (host) by executing firmware-level software
- DRAM: It is a volatile memory that provides faster read/write performance
- Host Interface: An SSD connects to the host machine using an interface. The commonly used SSD interfaces are SATA, PCIe, SCSI, etc.



Difference Between HDD and SSD



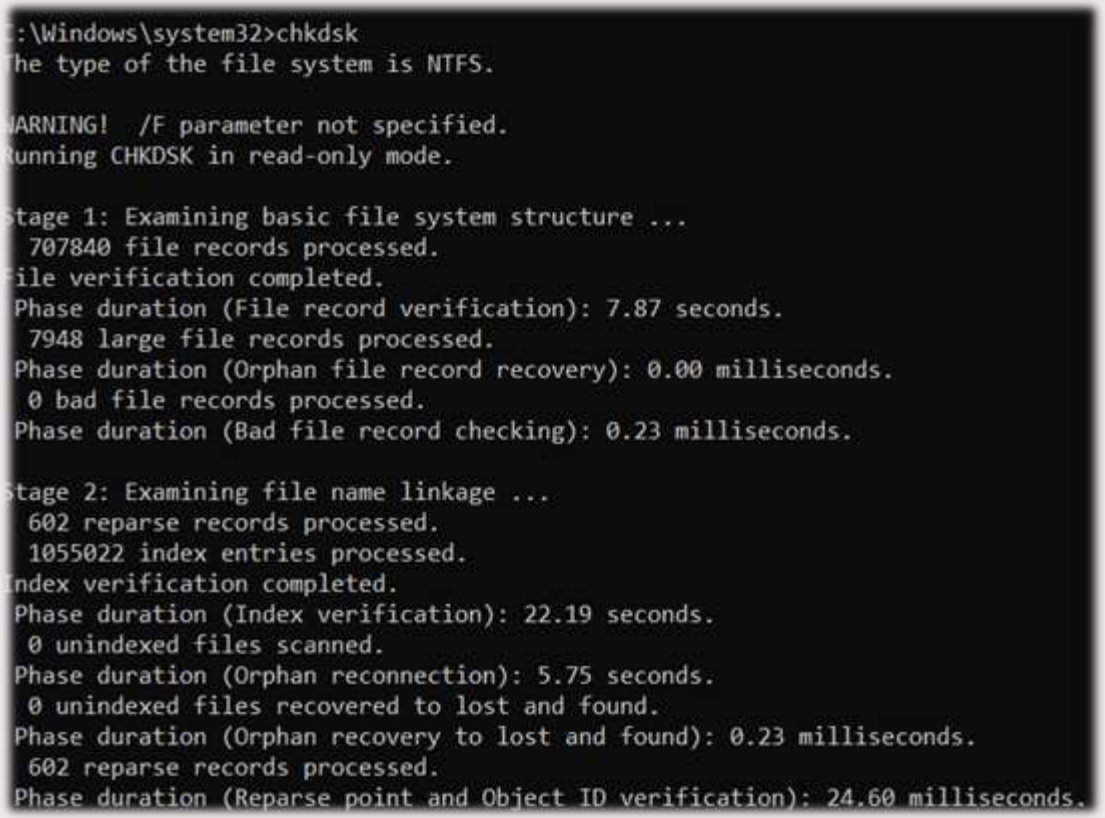
Hard Disk Drive (HDD)



Solid State Drive (SSD)

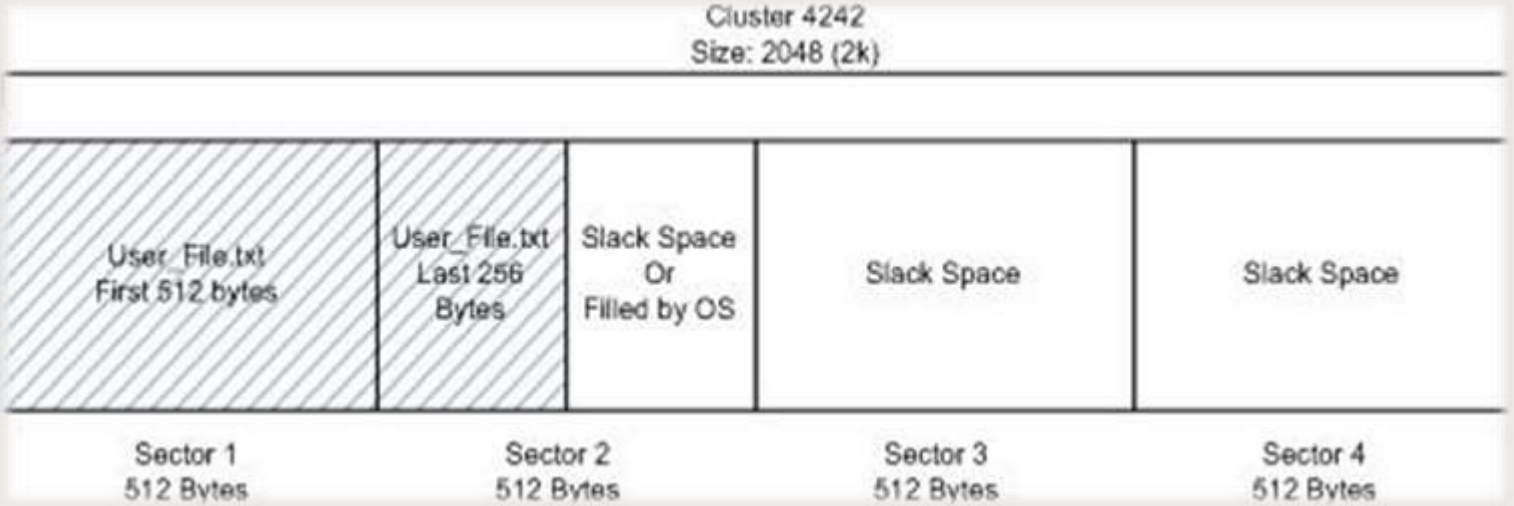
LOST CLUSTERS

- When the OS marks clusters as used but does not allocate them to any file, such clusters are known as lost clusters
- A lost cluster is a FAT file system error
- It is mainly the result of a logical structure error and not a physical disk error
- They usually occur because of interrupted file activities caused when, for example, a file is not properly closed; thus, the clusters involved in such activity are never linked correctly to a file
- **CHKDSK** is a system tool in Windows that authenticates the file system reliability of a volume and repairs logical file system errors



SLACK SPACE

- Slack space is the storage area of a disk between the end of a file and the end of a cluster If the file size is less than the cluster size, a full cluster is still assigned to that file. The remaining unused space is called slack space.
- For example, if the partition size is 4 GB, each cluster will be 32 KB in size. Even if a file requires only 10 KB, the entire 32 KB will be allocated to that file, resulting in 22 KB of slack space



MASTER BOOT RECORD (MBR)

- A master boot record (MBR) is the first sector ("sector zero") of a data storage device such as a hard disk
- The information regarding the files on the disk, their locations and sizes, and other important data is stored in the MBR file
- In practice, MBR almost always refers to the 512-byte boot sector (or partition sector) of a disk
- It is important in both **computer booting** and **digital forensics** because it contains information about how the disk is organized and, in BIOS-based systems, how the operating system starts.

Importance in Digital Forensics: During forensic examination, investigators may analyze the MBR to determine:

Partition information

Disk layout

Boot configuration

Evidence of boot-sector modification

Possible bootkits or boot-sector malware

Deleted or altered partition information

Original disk structure

GLOBALLY UNIQUE IDENTIFIER (GUID)

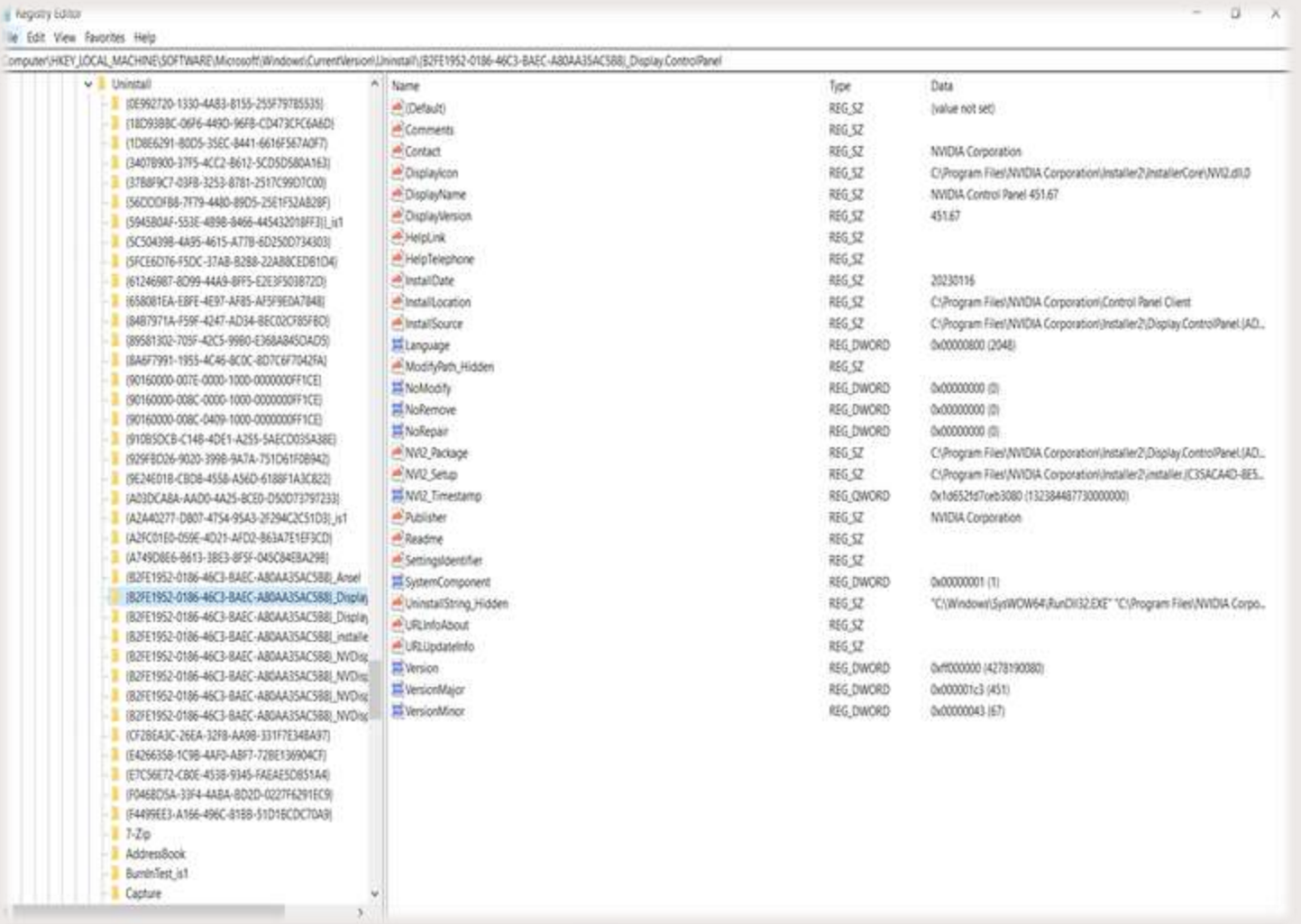
A **GUID (Globally Unique Identifier)** is a **128-bit value used to uniquely identify an object, device, partition, file, software component, or other resource**.

GUIDs are commonly represented as a **32-character hexadecimal value divided into five groups**.

GUID in Digital Forensics

In disk forensics, GUIDs are especially important with **GPT (GUID Partition Table)**.

Each GPT partition can have a **unique Partition GUID**, which helps identify that partition independently of its physical location or partition number.



FILE CARVING

- It is a technique to recover files and fragments of files from the hard disk in the absence of file system metadata
- It involves searching the raw data on a storage device for specific file signatures or patterns, and then extracting the data associated with those patterns
- A file header is a signature which is a constant numeric or text value that determines a file format
- Investigators can look at file headers to verify the file format using tools such as 010 Editor, Hex Viewer, WinHex, etc

It is especially useful when:

A file has been **deleted**

The file system is **corrupted**

File-system metadata is missing

A storage device has been **formatted**

The investigator is examining **unallocated space**

File fragments remain on a disk even though the directory entry is gone

IMPORTANCE OF FILE CARVING IN DIGITAL FORENSICS

- Recovers deleted files from unallocated space.
- Works without file-system metadata such as MFT or directory entries.
- Recovers evidence after formatting or file-system corruption.
- Identifies hidden or renamed files using file signatures.
- Extracts valuable evidence such as images, videos, PDFs, and documents.
- Useful for damaged storage devices and forensic disk images.
- Helps reconstruct digital activities during investigations.

FILE SYSTEMS

A file system is the method used by an operating system to organize, store, retrieve, and manage files on a storage device. It maintains information such as file names, locations, sizes, permissions, timestamps, and metadata.

1. **FAT — File Allocation Table**

FAT (File Allocation Table) is a file system used to organize and manage files on storage devices such as hard disks, USB drives, memory cards, and older computer systems.

It is called File Allocation Table because it maintains a table that keeps track of which storage clusters belong to which files.

Common versions:

FAT12

FAT16

FAT32

exFAT — newer version designed mainly for flash/removable storage

FAT FILE SYSTEM STRUCTURE

1. Boot Sector
(Reserved Area)
- BPB (Bios Parameter Block)
- File system parameters
- Boot code

(Sector 0)

2. FAT Region
- FAT #1
(File Allocation Table)

3. FAT Region
- FAT #2
(Backup of FAT #1)

4. Root Directory Region
(Fixed size in FAT12/16)
(In FAT32, it is in Data Area)

5. Data Area
(Contains file and directory data)

On Disk Layout

1. BOOT SECTOR (BPB – BIOS PARAMETER BLOCK)

Field	Description	Field	Description
Bytes per Sector	Usually 512, 1024, 2048 or 4096	Total Sectors	Total number of sectors
Sectors per Cluster	1, 2, 4, 8, 16, ...	Sectors per FAT	Size of one FAT
Reserved Sectors	Number of reserved sectors	Media Descriptor	Type of storage media
Number of FATs	Normally 2	Sectors per Track	Sectors per track
Root Dir Entries	For FAT12/16 (Fixed)	Heads / Cylinders	Geometry information

2. FAT (FILE ALLOCATION TABLE)

Cluster No.	FAT Entry (Next Cluster)	Meaning
0	Reserved	Reserved
1	FF8 – FFF	Reserved
2	000	Free Cluster
3	005	Next cluster is 5
...	...	
N	FFF	End of Cluster Chain
M	FF7	Bad Cluster

Example: Cluster Chain for a File

File: report.docx

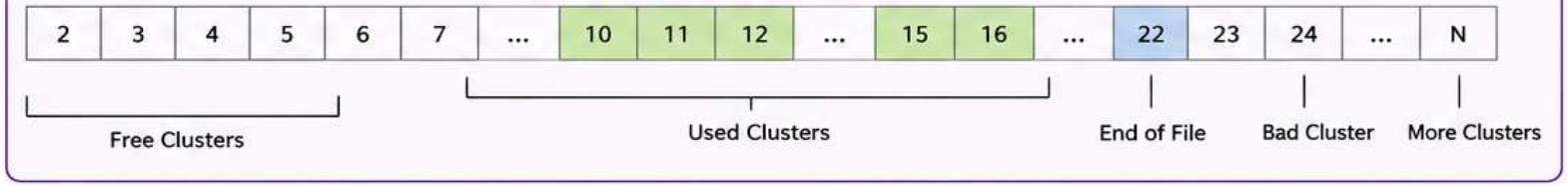
Cluster 10 → Cluster 15 → Cluster 16 → Cluster 22 → EOF (End of File)

FAT entries:
- 10 → 15
- 15 → 16
- 16 → 22
- 22 → EOF (End of File)

3. ROOT DIRECTORY / DIRECTORY ENTRIES

File Name	Ext	Attr.	Start Cluster	File Size	Date	Time
report	docx	20	10	24 KB	21-08-2026	10:15
photo	jpg	20	25	35 KB	20-08-2026	09:10
notes	txt	20	40	5 KB	19-08-2026	11:25
...	...	...	...	...	...	...

4. DATA AREA (CLUSTERS)



Free Cluster | Used Cluster | End of File Cluster | Bad Cluster

VERSIONS OF FAT FILE SYSTEM

1. FAT12

- **Introduced:** Early 1980s.
- Uses **12-bit** entries in the File Allocation Table.
- Mainly used for **floppy disks**.
- Supports relatively small storage devices.
- Common in early DOS-based computers.
- **Forensics:** Useful for investigating old floppy disks and legacy systems.

Partition/Volume Size: Approximately **16 MB**
Maximum File Size: Approximately **16 MB**
File Name: 8.3 format

2. FAT16

- Uses **16-bit** entries.
- Supports larger storage than FAT12.
- Used in **MS-DOS and early Windows systems**.
- Commonly used on older hard disks and removable media.
- Has a maximum partition size of about **2 GB** under the traditional implementation.
- **Forensics:** Directory entries, FAT tables, and unallocated clusters can contain recoverable evidence.
- **Partition/Volume Size:** Up to approximately **2 GB**
  Maximum File Size: Approximately **2 GB**
  File Name: 8.3 format originally

3. FAT32

- Introduced with **Windows 95 OSR2**.
- Uses **32-bit FAT entries** (though not all 32 bits are used for cluster addressing).
- Supports much larger partitions than FAT16.
- Commonly used in **USB drives, SD cards, and memory cards**.
- Maximum individual file size is approximately **4 GB**.
- **Forensics:** Deleted files can often be recovered from unallocated clusters if the data has not been overwritten.

Partition/Volume Size: Commonly up to **2 TB** in Windows implementations
Maximum File Size: 4 GB − 1 byte
File Name: Up to **255 characters** with Long File Name support.

4. exFAT

- Introduced by **Microsoft in 2006**.
- Designed mainly for **flash storage and large removable devices**.
- Supports very large files and volumes compared with FAT32.
- Commonly used for **USB flash drives and SDXC cards**.
- Has better support for modern large-capacity storage.
- Does **not use traditional FAT32-style journaling**.
- **Forensics:** Useful when investigating modern removable media.

Partition/Volume Size: Up to approximately **128 PB**
Maximum File Size: Up to approximately **16 EB − 1 byte**
File Name: Up to **255 characters**

NTFS — NEW TECHNOLOGY FILE SYSTEM

NTFS (New Technology File System) is the default file system used by modern Windows operating systems. It was introduced with **Windows NT** and provides better security, reliability, file management, and forensic information than FAT.

Basic NTFS Structure —

NTFS is mainly organized into these parts:

Boot Sector – Contains information about the NTFS volume, such as cluster size and MFT location.

MFT (Master File Table) – The **main database of NTFS**. It stores records and metadata for files and directories.

MFT Zone – Reserved space for future MFT growth.

System Files – Special files such as $LogFile, $Bitmap, $Boot, and $UsnJrnl that manage filesystem information.

Data Area – Contains the **actual contents of files**, stored in clusters.

NTFS

- **Partition/Volume Size**

Modern NTFS implementations can support volumes up to approximately **8 PB (Petabytes)**.

The actual usable limit depends on the Windows version and storage configuration.

- **Maximum File Size**
  - NTFS supports **very large files**, far beyond the **4 GB limit of FAT32**.
  - Modern implementations can support files up to approximately **8 PB** under suitable configurations.
- **Cluster Size**
  - A cluster is the basic storage unit used by NTFS.
  - **4 KB** is the commonly used/default cluster size.
  - NTFS supports cluster sizes up to **64 KB** for standard formatting.
- **File Name**
  - NTFS supports file names of up to **255 characters**.
  - It supports **Unicode**, allowing different languages and special characters.

DIFFERENCE BETWEEN FAT AND NTFS

Point	FAT	NTFS
1. Structure	Uses File Allocation Table (FAT)	Uses MFT (Master File Table)
2. File Size	FAT32: Maximum 4 GB − 1 byte	Supports very large files
3. Partition Size	FAT32: Commonly up to 2 TB in traditional Windows tools	Supports much larger volumes
4. Security	❌ No advanced file permissions	✅ ACL-based security
5. Journaling	❌ Not supported	✅ Supported
6. Reliability	Less reliable after crashes	More reliable due to journaling
7. Encryption	❌ No native EFS	✅ EFS supported
8. Compression	❌ No native compression	✅ Supported
9. Forensics	Less metadata available	Rich forensic metadata

APPLE FILE SYSTEMS

1. HFS — Hierarchical File System

- Developed by **Apple** for early Macintosh systems.
- Introduced in **1985**.
- Uses a hierarchical directory structure.
- Designed for older storage technologies.
- **Maximum file size:** approximately **2 GB**.
- Now considered **legacy/obsolete**.
- Mainly important for **historical digital forensics**.

2. HFS+ — Mac OS Extended

- Introduced in **1998** as an improved version of HFS.
- Used extensively by older versions of **macOS**.
- Supports **Unicode file names**.
- Supports larger files and volumes than HFS.
- Supports **journaling** in its journaled form.
- **Maximum file size:** approximately **8 EB theoretically**.
- Commonly encountered when examining **older Mac computers and drives**.
- Largely replaced by APFS.

APFS — APPLE FILE SYSTEM

- Introduced by Apple in **2016** and rolled out with **macOS 10.13 High Sierra in 2017**.
- It is the **modern Apple filesystem**.
- Designed especially for **SSDs and flash storage**.
- Used across modern Apple platforms, including **macOS, iOS, iPadOS, watchOS, tvOS, and visionOS**.
- Uses **64-bit inode/file identifiers** and supports very large volumes and files.
- Supports **Copy-on-Write (CoW)**.
- Supports **Snapshots**.
- Supports **Clones** of files/directories.
- Supports **Space Sharing** between APFS volumes.
- Supports strong **encryption**.
- Uses **checksums** for filesystem metadata.
- Provides improved performance and reliability compared with HFS+.

WINDOWS FORENSICS

Windows stores a large amount of information about system activity, user behavior, application execution, file access, devices, and security events. In digital forensics, these traces are called forensic artifacts.

The four important artifacts here are:

- Windows Registry
- Windows Event Logs
- Windows Prefetch
- Windows Shortcuts (LNK Files)